

Section 6.3: The Capabilities and Limitations of ICT

ICT devices and systems have many capabilities (**thinks they do well**) and limitations (**things they don't do so well**).

This section looks at some of these capabilities and limitations.

Key Concepts of this section:

- # Understand that ICT devices have both *capabilities* and *limitations*.
- # Be able to *discuss* some of these capabilities and limitations.

Capabilities of ICT devices

What things do ICT devices do well?

- # ICT devices and systems are capable of performing lots of tasks very well. Some of these are listed below:



1. Speed

Computers perform tasks very fast. Some **cpu's** can execute literally '**billions**' of instructions per second.

2. Accuracy

If programmed properly, computers are capable of performing the same **tasks 100% accurately every time**.

For example, a robot that is programmed to bolt a wheel to a car will get it exactly right each time.



Key Words:

Capabilities, Limitations

3. Multi-Tasking

Computers are able to perform **many tasks at the same time**.

4. Repetitive Tasks

Computers will carry out **boring and repetitive tasks accurately**. Humans tend to get bored and make mistakes.

For example, machines that count bank notes will do that job all day long without making a mistake.

5. Networking

Computers can be **joined together** (either as a LAN or a WAN) and allows them to easily **share resources** such as **printers** and **data**.

6. Huge amounts of Data Storage

Computer databases can store **enormous amounts of data**.

All types of data be stored including text, numbers, images and video.

7. Fast Searches

Computers can **search for stored data extremely fast**.

For example, the police can match fingerprints in a database in minutes rather than spending weeks searching through printed copies.

8. Security

Data can be **secured** with **passwords** or **encryption**.



Bank note counters perform repetitive tasks



Fingerprints can be matched quickly in a database



Programmed robots carry out jobs fast and accurately

What things do ICT devices not do well?

Computers, and other devices, are not without their problems. Limitations of ICT devices are listed below:



1. Battery Life

Portable device such as laptops, **PDAs** and **mobile phones** are **limited by their battery life**. These devices only remain charged for so long before they need to be plugged into the mains.



Laptop batteries have a limited charge

2. Weight

Some portable devices are **heavy** (especially laptops). This does not make them ideal for carrying around.

3. Screen and Keyboard Size

Smaller devices such as **mobile phones** and **PDAs** have very **small screens** and **keyboards** which can cause problems for some people.



Some devices have small screens and keyboards

4. Bandwidth

Bandwidth is the **speed that data can be transported over the Internet**.

Low bandwidth means that large files will take a long time to transfer.

Activity!

[Capabilities and Limitations of ICT - Questions](#)

Click the above task and follow the instructions given on the sheet. Make sure that you complete all questions in full.



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